ASTRO-NOT OBSERVATIONAL DATA PROJECT

Multi-Band Astronomical Catalog

Representative-Color Compositing from DSS2 Survey Archives

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Data Source: Second Digitized Sky Survey (DSS2) via NASA SkyView
Band Mapping: Blue (0.44μm) $\rightarrow$ B, Red (0.67μm) $\rightarrow$ G, Near-IR (0.85μm) $\rightarrow$ R
Dynamic Scaling: 99.5% Percentile Limits + Inverse Hyperbolic Sine (`asinh`)

Catalog Registry

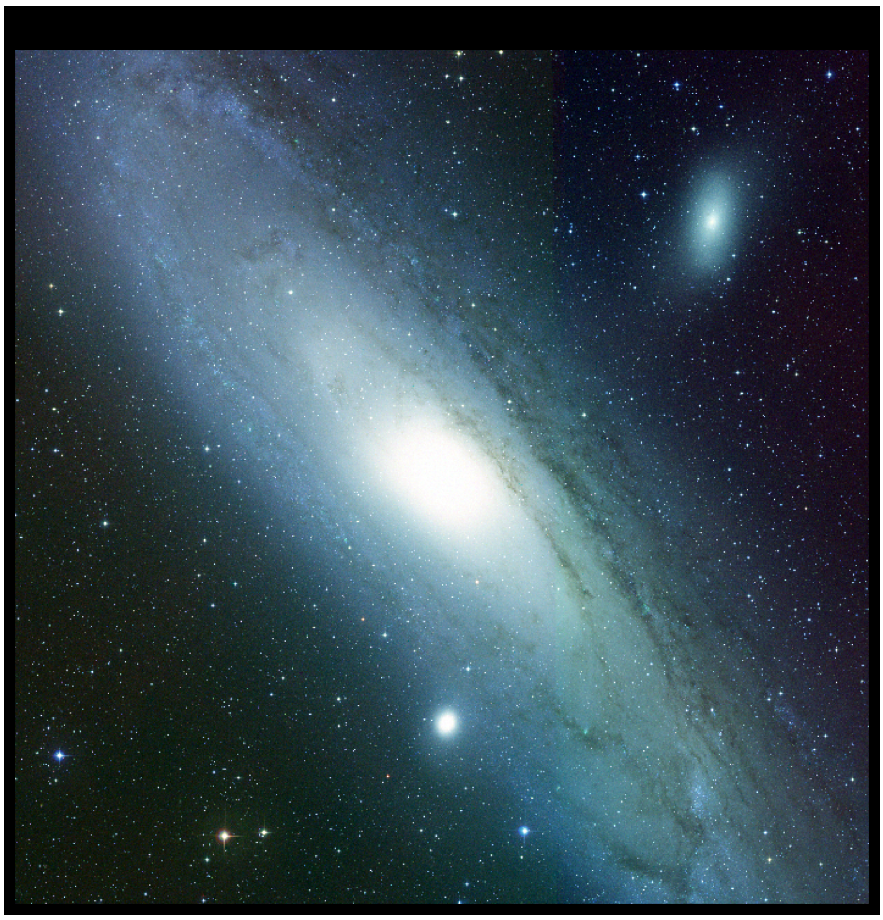
Catalog ID	Common Name	Object Classification	Constellation
M31 / NGC 224	Andromeda Galaxy	Barred Spiral Galaxy	Andromeda
M81 / NGC 3031	Bode's Galaxy	Grand Design Spiral Galaxy	Ursa Major
M1 / NGC 1952	Crab Nebula	Supernova Remnant / Pulsar Wind	Taurus
M27 / NGC 6853	Dumbbell Nebula	Planetary Nebula	Vulpecula
M15 / NGC 7078	Pegasus Globular Cluster	Core-Collapse Globular Cluster	Pegasus
M13 / NGC 6205	Hercules Globular Cluster	Globular Star Cluster	Hercules
M8 / NGC 6523	Lagoon Nebula	H II Emission Nebula / Open Cluster	Sagittarius
M42 / NGC 1976	Orion Nebula	Diffuse Emission / Reflection Nebula	Orion
M45 / Collinder 42	Pleiades Cluster	Open Cluster with Reflection Nebula	Taurus
M57 / NGC 6720	Ring Nebula	Planetary Nebula	Lyra
M104 / NGC 4594	Sombrero Galaxy	Unbarred Spiral / Elliptical Hybrid	Virgo
M33 / NGC 598	Triangulum Galaxy	Triangulum Spiral Galaxy	Triangulum
M20 / NGC 6514	Trifid Nebula	Emission / Reflection / Dark Nebula	Sagittarius
NGC 6992	Veil Nebula (Eastern)	Supernova Remnant Filament	Cygnus
M51 / NGC 5194	Whirlpool Galaxy	Interacting Grand Design Spiral	Canes Venatici

Data Acquisition Note: *While the primary multi-band compositing for this catalog leverages the Second Digitized Sky Survey (DSS2), inherent coverage gaps occasionally exist within the archival datasets for specific sky regions. Notably, for the Lagoon Nebula (M8) and Trifid Nebula (M20), DSS2 Blue-band observations were unavailable due to original survey constraints. To ensure complete spectral representation and maintain the structural integrity of the representative-color composites, archival Blue-band data from the First Digitized Sky Survey (DSS1) was systematically integrated for these two targets.*

M31 — Andromeda Galaxy

Spiral Galaxy

NGC 224 — Major Spiral Galaxy in the Local Group



Astrometric Metadata	Spectral & Processing Analysis
Distance: 2.5 million light-years	DSS2 Blue/Red/IR Band Mapping
Constellation: Andromeda	High-density stellar nucleus
Coordinates: RA 00h 42m 44s — Dec +41° 16’ 09”	Absorbing dust lanes and satellite dwarf galaxies

M81 — Bode's Galaxy

Spiral Galaxy

NGC 3031 — Grand Design Spiral Galaxy

Astrometric Metadata	Spectral & Processing Analysis
Distance: 11.8 million light-years	DSS2 Blue/Red/IR Band Mapping
Constellation: Ursa Major	Active Galactic Nucleus (AGN) core
Coordinates: RA 09h 55m 33s — Dec +69° 03' 55"	Grand design spiral arms with star formation

M1 — Crab Nebula

Supernova Remnant

NGC 1952 — Supernova Remnant of SN 1054

Astrometric Metadata	Spectral & Processing Analysis
Distance: 6,500 light-years	DSS2 Blue/Red/IR Band Mapping
Constellation: Taurus	Central Pulsar Wind Nebula power source
Coordinates: RA 05h 34m 31s — Dec +22° 00’ 52”	Complex ionized gas filaments

M27 — Dumbbell Nebula

Planetary Nebula

NGC 6853 — Bipolar Planetary Nebula

Astrometric Metadata	Spectral & Processing Analysis
Distance: 1,360 light-years	DSS2 Blue/Red/IR Band Mapping
Constellation: Vulpecula	Prolate gas envelope ejected by dying star
Coordinates: RA 19h 59m 36s — Dec +22° 43’ 16”	Ionizing white dwarf core

M15 — Globular Cluster

Star Cluster

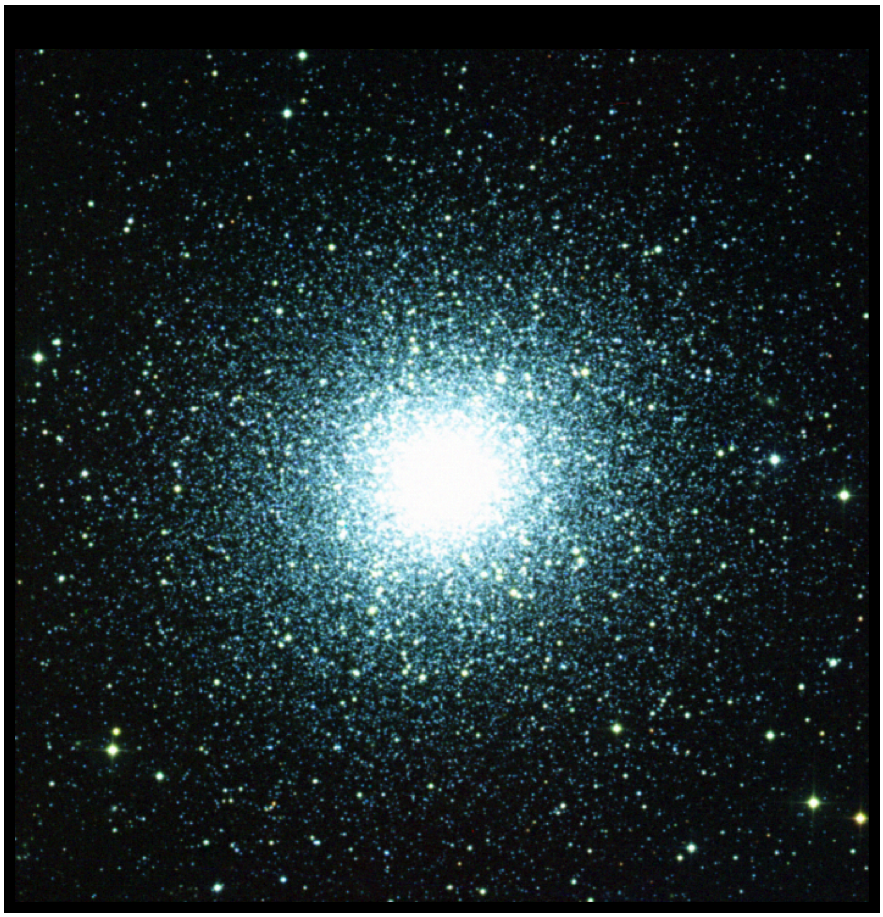
NGC 7078 — Dense Core-Collapse Globular Cluster

Astrometric Metadata	Spectral & Processing Analysis
Distance: 33,600 light-years	DSS2 Blue/Red/IR Band Mapping
Constellation: Pegasus	Gravitational core collapse structure
Coordinates: RA 21h 29m 58s — Dec +12° 10’ 01”	Over 100,000 Population II ancient stars

M13 — Hercules Globular Cluster

Star Cluster

NGC 6205 — Great Globular Cluster in Hercules



Astrometric Metadata	Spectral & Processing Analysis
Distance: 22,200 light-years	DSS2 Blue/Red/IR Band Mapping
Constellation: Hercules	Hundreds of thousands of bound stars
Coordinates: RA 16h 41m 41s — Dec +36° 27’ 35”	High central stellar core resolution

M8 — Lagoon Nebula

Emission Nebula

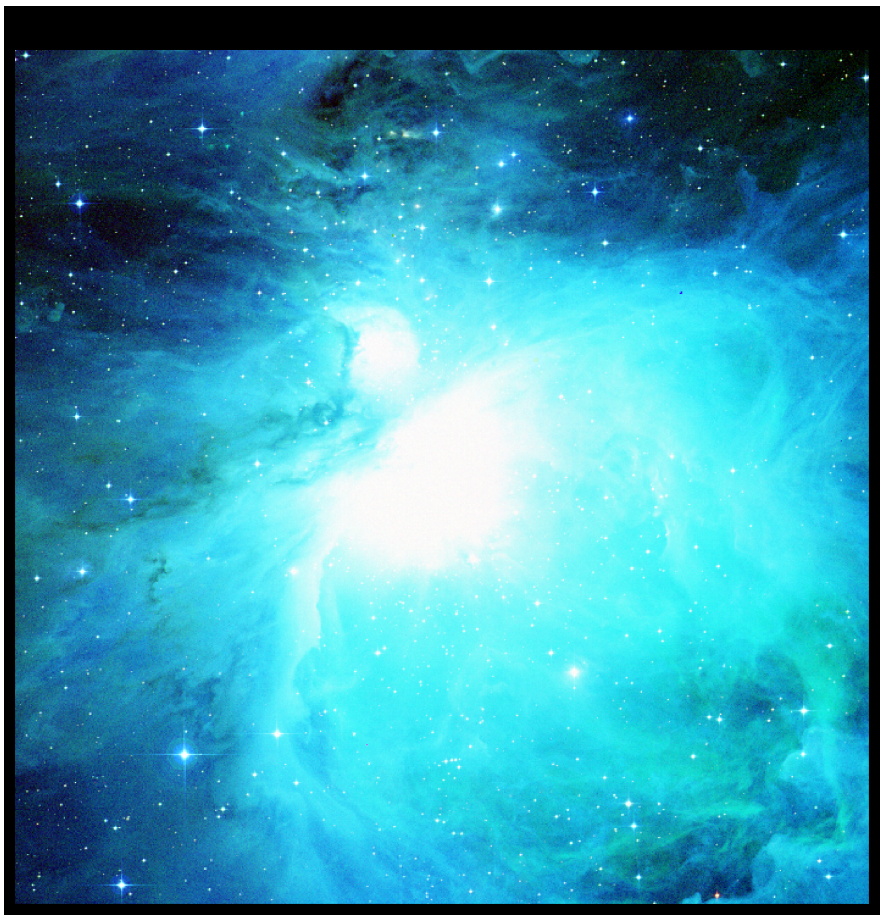
NGC 6523 — Massive H II Star-Forming Region

Astrometric Metadata	Spectral & Processing Analysis
Distance: 4,100 light-years	DSS1 (Blue) & DSS2 (Red/IR) Hybrid Mapping
Constellation: Sagittarius	Ionized H II hydrogen gas clouds
Coordinates: RA 18h 03m 37s — Dec -24° 23’ 12”	Embedded cluster NGC 6530 and Bok globules

M42 — Orion Nebula

Emission Nebula

NGC 1976 — Nearby Massive Star-Forming Region



Astrometric Metadata	Spectral & Processing Analysis
Distance: 1,344 light-years	DSS2 Blue/Red/IR Band Mapping
Constellation: Orion	Ionized by central Trapezium cluster stars
Coordinates: RA 05h 35m 17s — Dec -05° 23' 28"	`asinh` stretch preserves core and outer gas

M45 — Pleiades Cluster

Star Cluster

Melotte 22 — Open Cluster & Reflection Nebulosity

Astrometric Metadata	Spectral & Processing Analysis
Distance: 444 light-years	DSS2 Blue/Red/IR Band Mapping
Constellation: Taurus	Hot B-type luminous blue stars
Coordinates: RA 03h 47m 24s — Dec +24° 07’ 00”	Reflection nebulosity from interstellar dust

M57 — Ring Nebula

Planetary Nebula

NGC 6720 — Shell-Like Planetary Nebula

Astrometric Metadata	Spectral & Processing Analysis
Distance: 2,570 light-years	DSS2 Blue/Red/IR Band Mapping
Constellation: Lyra	Barrel-shaped expanding ionized gas shell
Coordinates: RA 18h 53m 35s — Dec +33° 01’ 45”	Central white dwarf star remnant

M104 — Sombrero Galaxy

Spiral Galaxy

NGC 4594 — Unbarred Spiral Galaxy with Bulge

Astrometric Metadata	Spectral & Processing Analysis
Distance: 31.1 million light-years	DSS2 Blue/Red/IR Band Mapping
Constellation: Virgo	Prominent dark dust lane in thin disk
Coordinates: RA 12h 39m 59s — Dec -11° 37’ 23”	Unusually large and bright central stellar bulge

M33 — Triangulum Galaxy

Spiral Galaxy

NGC 598 — Local Group Spiral Galaxy

Astrometric Metadata	Spectral & Processing Analysis
Distance: 2.73 million light-years	DSS2 Blue/Red/IR Band Mapping
Constellation: Triangulum	Loose spiral arm structure with ionized H II regions
Coordinates: RA 01h 33m 50s — Dec +30° 39’ 37”	Third largest member of the Local Group

M20 — Trifid Nebula

Diffuse Nebula

NGC 6514 — Combination Emission/Reflection Nebula

Astrometric Metadata	Spectral & Processing Analysis
Distance: 4,100 light-years	DSS1 (Blue) & DSS2 (Red/IR) Hybrid Mapping
Constellation: Sagittarius	Tri-lobed structure separated by dark dust lanes
Coordinates: RA 18h 02m 23s — Dec -23° 01’ 48”	Active star-forming H II complex

NGC 6992 — Veil Nebula

Supernova Remnant

Eastern Veil Nebula — Cygnus Loop Fragment

Astrometric Metadata	Spectral & Processing Analysis
Distance: 2,400 light-years	DSS2 Blue/Red/IR Band Mapping
Constellation: Cygnus	Fine filamentary ionized shock waves
Coordinates: RA 20h 56m 20s — Dec +31° 43’ 00”	Remnant of ancient massive star collapse

M51 — Whirlpool Galaxy

Spiral Galaxy

NGC 5194 — Interacting Grand Design Spiral

Astrometric Metadata	Spectral & Processing Analysis
Distance: 31 million light-years	DSS2 Blue/Red/IR Band Mapping
Constellation: Canes Venatici	Gravitational interaction with companion galaxy NGC 51
Coordinates: RA 13h 29m 52s — Dec +47° 11’ 43”	Well-defined spiral arms with active starburst zones